Sam Naveenkumar V

Bachelor of Technology
Artificial Intelligence and Data Science
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Education

Degree/Certificate	Institute/Board	Year
B.Tech. (AI & DS)	Karunya Institute of Technology and Sciences, Coimbatore	2022-2026
ISC(XII)	Vikaasa Heritage Campus, Madurai	2022
ISCE(X)	Vikaasa Heritage Campus, Madurai	2020

Experience

- **AI Research Intern (Generative AI)** *May 2025 - July 2025*
  BluBridge Technologies (LLM Research Lab) Chennai, India, (On-site)
  – Built ZNN, a Deep Learning autograd (Pytorch) engine from scratch with FFNN, Backprop, base of LLM pretraining.
  – Built an Autoregressive LLM from scratch using GPT-2 transformer architecture with BPE tokenizer, FFNN, Multi-Head Self-Attention (Flash Attention), Layer Normalization, Dropout, AdamW optimizer and LoRA fine-tuned.
- **Product Development Engineer Intern (AI & Computer Vision)** *June 2024 - July 2024*
  A&T Video Networks (OEM Company) Madurai, India (On-site)
  – Developed AI solutions: face recognition, autoframing, and object tracking using YOLOv11 and OpenCV.
  – Built a smart attendance system with Vision-Language Models (VLM), ML pipelines and DeepFace integration.
  – Optimized camera drivers and enhanced AI features in collaboration with Aver (Taiwan).

Projects

- **Rachel, the AI-HR Interview Coach**
  Multimodal AI HR interview platform generating domain-specific questions from experience, projects, and skills in candidate's resume. Includes AI video & audio proctoring (speech, head pose, tab switch) and real-time response evaluation for preparation.
  – **Tools & technologies**: NLP, Vision AI, GenAI, RAG, NVIDIA NeMo, Guardrails, LangChain, OpenWeb UI, LangGraph
- **DeCAPTCHA**
  Cracked Google reCAPTCHA V2 using a GAN Deep Learning pipeline where CNNs and OCR extract text (generator) and Vision-Language Model(VLM) verify patterns (critic). Implemented real-time image segmentation and adversarial refinement.
  – **Tools & Technologies**: TensorFlow, Keras, OpenCV, Tesseract OCR, CUDA, Docker, Kubernetes, React.js, FastAPI.
- **Multilingual Translation Engine (MTE)**
  Built a real-time low-latency multilingual translation platform using ASR, STT, NMT, and TTS for all 22 Indian languages. Enabled end-to-end text and voice interaction with speaker diarization, regional accents, dialects, and script variations.
  – **Tools & Technologies**: PyTorch, Hugging Face HF Transformers, NVIDIA NeMo, LangChain, FastAPI, Langserve, NVIDIA Riva, Triton Inference Server, MilvusDB, Docker, RAG, Vector Embeddings, Real-time Inference, Ansible.
- **CvNova (AI Resume Optimizer)**
  Agentic AI system with self-critiquing "Generator–Critic–Refiner" loop using Constitutional AI and Chain-of-Thought (CoT) prompt chaining AI. Automates resume optimization with ATS scoring, skill–role alignment, and refined job-matching outputs.
  – **Tools & Technologies**: OpenAI/Gemini LLMs, MoE, Pinecone, FAISS, FastAPI, Streamlit, GitHub DevOps, AWS.

Technical Skills

- **Programming:** C, C++, Python, Java, JavaScript, SQL, PHP, Redux, Linux OS(Ubuntu), WordPress, Pytest
- **Full Stack Dev:** HTML, Tailwind CSS,MongoDB,Express.js, ReactJS, AngularJS,Node.js, Svelte, Flutter, KubeFlow
- **DevOps:** Kubernetes, Jenkins, Terraform, Postman, Selenium, DevSecOps, Webhook, GCP, AWS, Azure, AutoKeras
- **GenAI Skills:** LLM Fine-Tuning, Pretraining, Model Quantization Distillation, Transformers, Human-in-the-loop, MCP, DeepEval, HumanEval, Prompt injection, LLM Jailbreak, NIMS, SLM, Prompt Engineering, CoT, XAI, vLLM, A2A, ACP, Embeddings, Vector DBs, Multimodal AI, Inference Optimization, PEFT (LoRA, QLoRA), RLHF, AI Agents

Positions of Responsibility

–**Innovator:** Karunya Technology Business Incubation Park (K-TBIP), Jul. 2023 – Present
–**Coordinator:** Generative AI Club, Karunya Innovation and Design Studio (KIDS), KITS, Jan. 2024 – Present
–**Trainer:** Karunya Innovation and Design Studio (KIDS), Coimbatore, Jul. 2023 – Present
–**Department Representative:** Dept. of AI, Data Science Cyber Security, KITS, Jul. 2025 – Present

Achievements

- **Secured 1st Position** LEOFEST'24, National Paper Presentation at Bannari Amman Institute of Technology, Erode
- **Secured 2nd Place** BARNSTORMGZ 2023, National Level Symposium,Hindustan University, Coimbatore
- **Finalist** 6th INAE - SERB - Youth Conclave GITAM, Visakhapatnam Andhra Pradesh 2023

Certifications

- Coursera Google AI Essentials, Cybersecurity, Digital Marketing & E-Commerce
- NPTEL Certification on Ethical Hacking, Cloud Computing, Technical English for Engineers.
- Microsoft Azure AI-900 Certification, Career Essentials in Generative AI by Microsoft and LinkedIn
- IBM Artificial Intelligence Fundamentals, Machine Learning Methods and Tools,